Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/31/2006
PI Name:	Sukharev, Sergei Ph.D.		
Project Title:	Mechanosensitive Channels In Bacteria: The Role Of Protein-Lipid Interactions In The Mechanism Of Gating		
Division Name:	Space Biology		
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Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
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Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
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No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
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Contact Monitor:		**Contact Phone:**	
Contact Email:			
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Task Description:	Today, we are only approaching a minimal level of understanding of how cells may sense and respond to mechanical stimuli. There is a drastic mismatch between the 'functional' phenomenology collected at the organismal and cellular levels and the molecular description of mechanosensory mechanisms. MscL, a mechanosensitive channel of large conductance, was the first isolated molecule shown to respond to membrane stretch by opening a large aqueous pore. It resides in the inner membrane of Escherichia coli and serves as protective safety valve in the event of osmotic shock. The channel is a pentameric complex made of 15 kDa subunits and responds directly to the tension in the lipid bilayer. During the previous funding periods, we determined the overall structure of MscL, and evaluated energetic and spatial parameters that characterize conformational transitions in native MscL. The crystal structure of MscL homolog now guides the research towards an atomic-scale description of the gating process. The analysis of mutations has been narrowed to functionally important regions such as short N-terminal helix, two transmembrane domains of the protein,		

	and the loop between them. The proposed study will be focused on identification of individual or groups of residues that determine the energy of closed-to-open transitions and the dwell time in each state. The methodology includes site-directed mutagenesis, covalent protein modifications, patch- clamp measurements, thermodynamic and kinetic analysis of dose-response curves and rates of transitions, videomicroscopy, molecular modeling and simulations. The study of MscL, a highly convenient model system, will give us basic understanding and the first example of what type of intramolecular interactions, sequence motifs, and conformations make membrane proteins sensitive to a physiologically relevant mechanical stimulation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research leads to understanding of very basic mechanisms of force detection by macromolecules in living organisms. It will help to clarify general principles and mechanisms of mechanosensation, which encompass a wide range of phenomena from simple osmoregulation in bacteria to complex phenomena such as gravitropism in plants and balance and hearing in humans. MscL represents simple and powerful model system highly useful for molecular and biophysical experiments as well as for molecular computation. Obtaining a complete and self-consistent picture of mechanosensitive channel gating which includes the protein, surrounding lipids and solvent would be a stepping stone to understanding the mechanisms of the growing families of sensory channels modulated by membrane stretch cell volume, lipids, amphipathic substances, or temperature.
Task Progress:	An important trait of the model of the MscL gating transition is an iris-like expansion of the transmembrane part of the protein complex. The five pairs of transmembrane helices forming the barrel are predicted to tilt substantially while moving away from the axis of the pore. During the past year we obtained new data on the character of MscL-lipid interactions that permits helical tilting. Aromatic residues, tyrosines or tryptophanes that anchor both ends of transmembrane helices in many integral proteins to membrane interfaces are not common in MscLs. This characteristic may be vital to MscL function as strong helical reorientation is necessary to open the channel. A single tyrosine (Y75) on the extracellular side of EcoMscL (one per subunit) is absent in TbMscL, which instead has a single tyrosine (Y87) on its cytoplasmic side. Moving the tyrosine of EcoMscL to the intracellular side (Y75F/F93Y) or capping the M2 helix on both sides (F93Y/W) increases the mean open time and the threshold for activation leading to a partial loss-of-function in osmotic shock survival assays. Increasing the distance between the caps by two helical turns (L102Y) apparently removes restraints for tilting and partially restores function. Capping the M2 helix with a charged residue (Y75E) makes the channel ‘stiff’ and abolishes function. Introducing an aromatic ‘cap’ into the M1 helix (I41W) decreases the activation threshold and shortens the mean open time, but unexpectedly leads to a complete loss-of-function in vivo. This data shows that restraining helical positions in MscL by introducing specific protein-lipid interactions at membrane interfaces compromises MscL function. There appears to be no simple correlation between the activation threshold and gain- or loss-of-function phenotypes.
Bibliography Type:	Description: (Last Updated: 03/01/2007)
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